ALBERTA DISTANCE LEARNING CENTRE

SSN2171

Personal Psychology 20 KEY

Assignment Booklet 7

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Summary

	Possible Marks	Your Marks	%
Part One	**22**		
Part Two	**10**		
Part Three	**10**		
Part Four	**25**		
TOTAL	**67**		

Teacher's Comments

Teacher's Signature

CANADIAN CATALOGUING IN PUBLICATION DATA

SSN2171
Personal Psychology 20
Assignment Booklet Package

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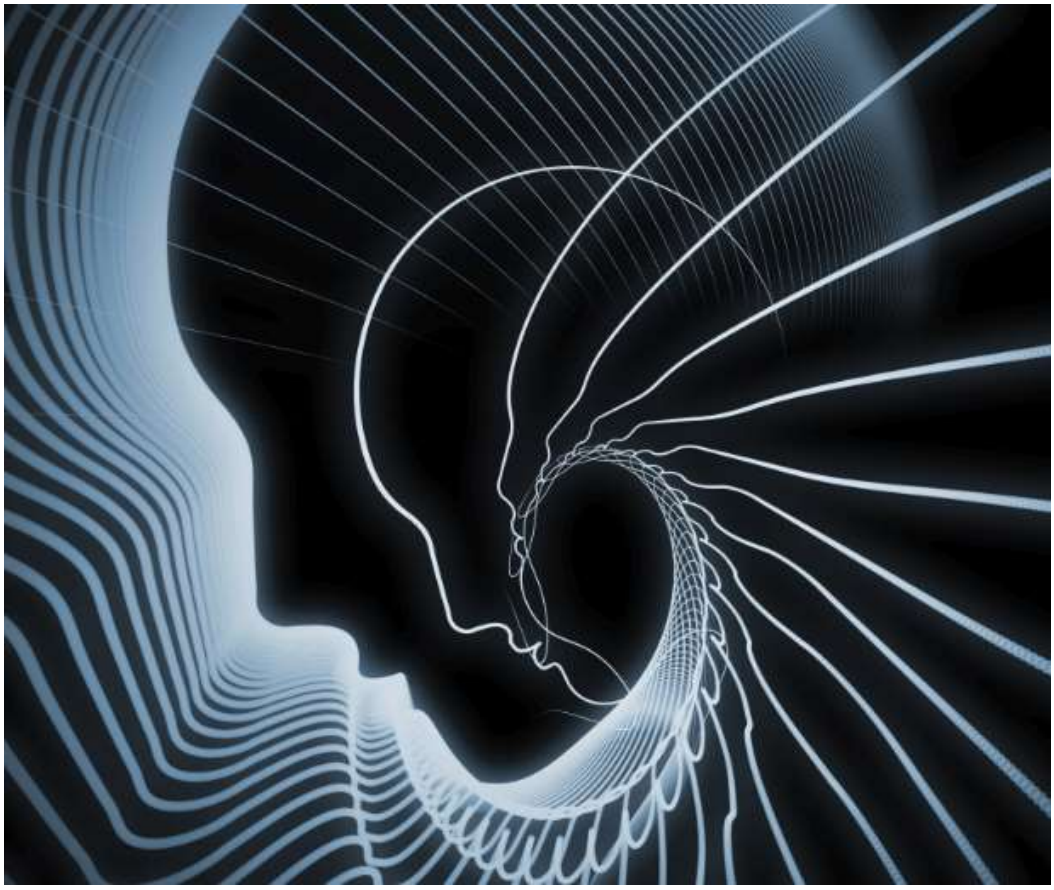
Personal Psychology 20



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Assignment Booklet 7

Instructions for Submitting Assignment Booklets

1. Submit Assignment Booklets regularly for correction.
2. **Submit only one Assignment Booklet at one time.** This allows your teacher to provide helpful comments that you can apply to subsequent course work and exams (if applicable).
3. **Check the following** before submitting each Assignment Booklet:
   a. Are all assignments complete?
   b. Have you edited your work to ensure accuracy of information and details?
   c. Have you proofread your work to ensure correct grammar, spelling, and punctuation?
   d. Did you complete the Assignment Booklet cover and attach the correct label?

Postal Mail

Determine sufficient postage by having the envelope weighed at a post office. (Envelopes less than two centimetres thick receive the most economical rate.)

Alternate Methods

Contact your teacher if you require an alternate method for submission of assignments.

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Assignment Booklet 7

Assessment

For successful completion of this course, you must

1. **Complete all questions in each Assignment Booklet to the best of your ability.**

2. **Achieve at least 40% on the final exam.**

3. **Achieve a final course mark of at least 50%.**

Process

- **Read the course material** before working on the activities in this Assignment Booklet. The Module Book contains the information necessary for successful completion of your work.

- **Proceed carefully** through each assignment. Reflect upon your answers and prepare your written responses to communicate your thoughts effectively. Time spent in planning results in better writing.

- **Proofread your work** before submitting it for marking. Check for content, organization, paragraph construction (if applicable), grammar, spelling, and punctuation.

- **If you encounter difficulties or have any questions**, contact your course teacher at Alberta Distance Learning Centre for assistance.

Format

If you choose to **use a word processor** for your written work,

- format your work using an easy-to-read 12-point serif font such as Times New Roman
- include your full name and student file number as a document header
- double-space your final copy
- staple your printed work to this Assignment Booklet

Avoid plagiarism by acknowledging all sources you use. Contact your teacher if you are uncertain how to document sources.

Although you are encouraged to work collaboratively and discuss various aspects of this course with others, **all submitted work must be your own**.

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Value 22

Part One: Multiple-Choice Items

Read each item carefully and decide which choice best completes the statement or answers the question. Write the letter of your choice in the blank before the item.

Point of clarification regarding sensation and perception:

Sensation refers to the process of receiving stimuli from the environment using one of the five senses and then having that information forwarded to our brains for processing. Perception is the processing and interpretation of the sensory stimuli we receive to better understand our environment.

Use the following information to answer multiple-choice items 1 and 2.

Robert loves vegetables and after he eats his bowl of steamed broccoli with fresh butter and sea salt, he decides that broccoli is his new favourite vegetable.

_B__ 1. As Robert places a broccoli floret in his mouth, breathes in deeply, and begins to chew, he is experiencing a(n)

A. illusion
B. sensation
C. perception
D. hallucination

_C__ 2. When Robert states that broccoli is his new favourite vegetable, this is due to his

A. illusion
B. sensation
C. perception
D. hallucination

_C__ 3. ESP is

p112

A. the sixth sense
B. a visual illusion
C. not scientifically proven
D. something everyone has

_B__ 4. Eyes are designed to detect

p115

A. colour
B. light waves
C. features of objects
D. electrical impulses

_A__ 5. The pupil

p112

A. expands in dim light
B. contracts in dim light
C. enlarges in bright light
D. does not change shape

_A__ 6. The coloured part of the eye is called the

p112

A. iris
B. lens
C. pupil
D. cornea

_C__ 7. The dark centre of the eye is called the

p112

A. iris
B. lens
C. pupil
D. cornea

_D__ 8. The transparent covering at the front of the eye is called the

p112

A. iris
B. lens
C. pupil
D. cornea

_A__ 9. This part of the eye controls the size of the pupil

p112

A. iris
B. lens
C. cornea
D. ciliary muscles

_B__ 10. This part of the eye focuses the images we see

p113

A. iris
B. lens
C. pupil
D. cornea

__C_ 11. One colour is distinguished from another colour by its

p115

A. power
B. intensity
C. wavelength
D. feature detectors

_D__ 12. Specialized cells called _____ are concentrated mainly in the centre of the retina and work _____.

p114

A. rods; in dim light
B. cones; in dim light
C. rods; to detect colour
D. cones; to detect colour

__A_ 13. Specialized cells called _____ are concentrated mainly around the edge of the retina and work _____.

p114

A. rods; in dim light
B. cones; in dim light
C. rods; to detect colour
D. cones; to detect colour

_D__ 14. The blind spot is caused by

p114

A. the iris closing
B. colour blindness
C. a concentration of photoreceptors
D. the space where the optic nerve leaves the eye

Point of clarification regarding cellular layers in the retina:

Page 113 incorrectly states that light first goes through the photoreceptor layer (rods and cones) then the bipolar layer and finally the ganglion cell layer. What actually happens is that light entering the eye passes through the ganglion cell layer then the plexiform and nuclear layers (with bipolar cells) on its way to the photoreceptor layer where the rods and cones are located. The rods and cones then activate and send *signals* **back** to the ganglion cells through the bipolar cells.

__D_ 15. Individuals with monochromatism see only in _____ because their _____ do not function.

p115

A. the three primary colours; rods
B. the three primary colours; cones
C. shades of white, grey, and black; rods
D. shades of white, grey, and black; cones

_C__ 16. Individuals with protanopia perceive only shades of

A. red and blue
p116 B. red and yellow
C. blue and yellow
D. white and black

_A__ 17. Which set of colours are complementary?

p117

A. red and green
B. blue and green
C. yellow and blue
D. purple and orange

_C__ 18. When you mix two primary colours you get a

p117

A. tint
B. shade
C. secondary colour
D. intermediate colour

_D__ 19. The cochlea helps to transmit vibration into neural impulses by

p125

A. detecting features
B. absorbing sound waves
C. stimulating the auditory nerve
D. causing hairs on its surface to vibrate

_C__ 20. Pheromones

p127

A. are hormones
B. have strong odours
C. signal sexual receptivity
D. are only active in insects

_C__ 21. The placebo effect occurs when people report feeling

p128

A. less pain when they have taken real medication
B. more pain when they have taken real medication
C. less pain when they have taken inactive (fake) medication
D. more pain when they have taken inactive (fake) medication

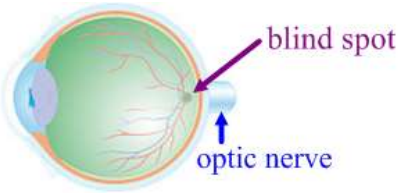
_C__ 22. In the wild, _____ is considered the most important sense.

p116

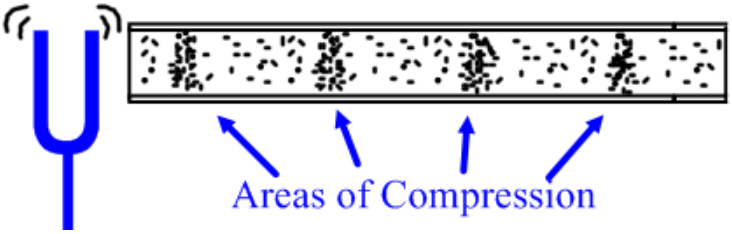
A. smell
B. touch
C. vision
D. hearing

Corrections to Course Content Regarding the Five Senses

- **Sight:** The diagram on page 113 incorrectly indicates that the blind spot is located in the vitreous humour at the back of the eye just in front of the retina. The blind spot is actually located where the optic nerve enters/exits the retina.



- **Sound:** The diagram on page 125 depicts sound waves as transverse waves (like light waves) with high and low points (crests and troughs) in the waves. Sound, in fact, is a longitudinal wave. A correct diagram of longitudinal waves formed when a tuning fork is struck near a pipe with air in it is as follows:



  - The distance between two consecutive areas of compression (wave cycles) determines the pitch of the sound wave. Long distances between compressions result a lower pitch (lower frequency). Shorter distances between compression result in a higher pitch (higher frequency). *Frequency* is the number of compressions (wave cycles) in a given time period.

  - The number of air molecules disturbed or compacted in a compression area relates to the sound pressure of a sound wave. The pressure (also known as *amplitude*) relates to the loudness or quietness of a sound. Without molecules (air, water, etc.) to transmit the energy (pressure) of a sound wave, there is no audible sound. In other words, if a hammer strikes a metal bar in a container that has had all the air removed (i.e., a vacuum), you would not hear anything.

- **Taste & Smell:** The course materials correctly identify that taste and smell are intimately entwined and function together to enhance the experience of eating. On page 126, however, the materials identify four basic tastes: sour, salty, sweet, and bitter. There is actually a fifth basic taste called umami. The word umami is borrowed from the Japanese culture and is translated as "pleasant savory taste." Ripe tomatoes are thought to be rich in umami flavour!

- **Touch:** This sense includes pain, temperature (hot and cold), as well as pressure. There are two typographical errors on page 128 that substitute "pleasure" for "pressure." Please note the correct term is pressure.

Value 10

Part Two: Matching – The Auditory System

Choose the letter of the term that best describes the concept or completes the sentence. Terms may not be used more than once.

p124-126 for all

	Statements	Terms
_A__	1. This is term refers to the strength, pressure, or "loudness" of a sound wave.	A. amplitude
_H__	2. This is the outer shell of the ear.	B. anvil
_D__	3. This structure transfers auditory information to the brain.	C. auditory cortex
_E__	4. This curled tube-like structure contains fluid.	D. auditory nerve
_G__	5. This is the first of three tiny bones in the inner ear that relay vibrations from the external environment.	E. cochlea
_F__	6. This refers to the number of compressions in a given time period.	F. frequency
_B__	7. Vibrations from the hammer are relayed to this tiny bone in the inner ear.	G. hammer
_C__	8. This area receives and translates auditory information.	H. pinna
_J__	9. This blocks and protects the delicate bones of the inner ear from the external world.	I. stirrup
_I__	10. This structure transfers vibrations to the cochlea.	J. tympanic membrane

Value 10

Part Three: True or False

Read each item carefully and decide if the statement is true or false. Write **T** or **F** in the blanks to indicate your choices.

_T__ 1. Colour blindness is genetic.
p115

_F___ 2. All animals have colour vision because colour is used to attract mates.
p116

_T__ 3. A tertiary colour is a combination of a primary colour and a secondary colour.
p117

_T__ 4. People with hearing loss may not realize their hearing has deteriorated.
p124

_F__ 5. Taste buds are only located on the edges of the tongue but taste buds respond to all tastes to varying degrees.
p126

_T__ 6. People with brain injuries that result in a loss of smell may then also experience a loss of taste.
p127

__T__ 7. Some parts of our skin have more sensory touch receptors than other parts of our skin.
p128

__T__ 8. Water droplets can act like prisms and separate light into many colours.
p119

__F__ 9. Humans attend to sound only in one ear.
p130

_F_ 10. When you add black to a colour you get a tint.
p117

Value 25

Part Four: Written-Response Items

Respond to the following items on the blanks below the items.

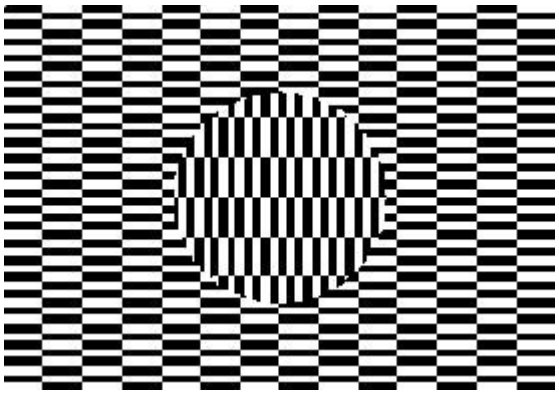
Addition Information Regarding Optical Illusions

Mechanisms that create optical illusions are located in various places along the visual pathways. Some of these mechanisms come from the retina itself, but most result from the way that the images captured with the eyes are reconstructed by the visual cortex.

The information our brain receives from our eyes doesn't always accurately represent reality. For example, some of the information received from the 125 million photoreceptors is lost as the information flows to 100 times fewer ganglion cells before being transmitted to the brain by the optic nerve.

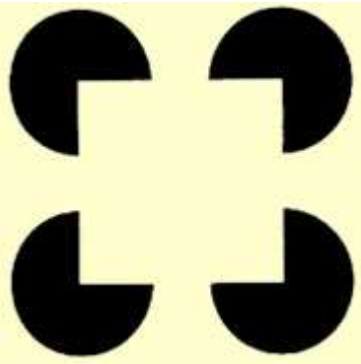
To compensate for the loss of information, the brain introduces abstract factors that complete or amplify the components of reality that are transmitted. In doing so, the brain sometimes creates an impression of unity/logic where there is none—in other words, an optical illusion. There are many types of optical illusions including those related to depth, colour, motion, and size. The written-response items 1 and 2 relate to optical illusions.

2 1. Below is an optical illusion. Hold your head still and move this page to the left then right and describe what you see.



Students should indicate this is a movement illusion… so the lines inside the circle should appear to move.

2 2. Describe what you see in the optical illusion below.



This is a geometric illusion so students should state they see a square. In reality, there is no square. (Because the missing quarters of the four discs all face inward, they create the perception of a square with a subjective contour.)

1 3. What is the name of the taste sensation (basic taste) with a Japanese name?

Umami

2 4. What happens if you plug your nose and eat an apple?

p127 *Smell is essential to experience taste so if your nasal passages are blocked, you cannot taste.*

2 5. How does the mind compensate when experiencing sensory deprivation?

p131 *The mind compensates by creating its own stimuli (e.g., hallucinations).*

2 6. How does the mind compensate when experiencing sensory overload?

p131 *The mind compensates by selectively attending to only certain stimuli (narrow focus).*

14 7. Select **one** of the following two options on the following pages and complete the associated tasks. Remember to enumerate (number) your responses on the response page for Option A so your marker knows the item to which you are responding. Responses for Option B may be provided within the Option itself.

Option A: Hot and Cold

As you have learned, the cutaneous sense (touch) involves pressure, pain, and temperature. For this option you may need to conduct internet (or other) research and you will conduct a short experiment regarding temperature. You will need three tall glasses (identical glasses if possible), water of varying temperatures, and a clock/watch/timer to time yourself. Once you have the materials, complete the following steps in the order they appear then answer the written-response items that follow in the space provided on page 14:

Steps:

- Fill one glass with ice water (very cold), fill one glass with room-temperature water (you can leave the glass of water on a table for a few hours to reach room temperature), and fill one glass with very hot water (not boiling).
- Grasp the glass of hot water with one hand, making sure the palm of your hand is touching the glass.
- Grasp the glass of ice water with your other hand, making sure the palm of your hand is touching the glass.
- Hold the glasses (hot water in one hand, ice water in the other) for 60 seconds.
- After 60 seconds, grasp the room-temperature glass with both hands, palms touching the glass.

1. Describe, with some detail, the sensations of touching the room-temperature glass after holding the other two glasses for 60 seconds. (4 marks)

2. What is the commonly used temperature scale in Canada called? (1 mark)

3. What is the commonly used temperature scale in the United States called? (1 mark)

4. What is the average body temperature of an adult in the temperature scale used in Canada? (2 marks)

5. Assessing a patient's temperature is part of a full physical examination. Besides taking a rectal (anal) reading of temperature, list two other sites that may be used for measuring temperature. (2 marks)

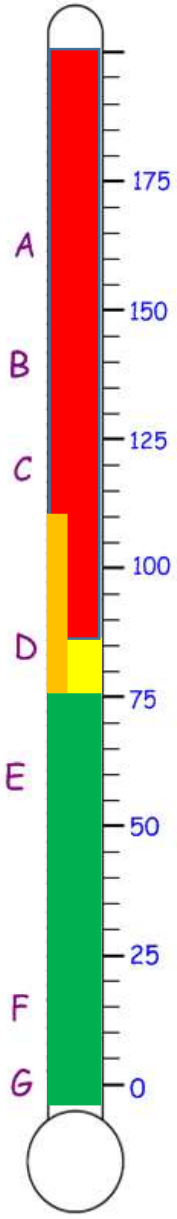
6. Body temperature fluctuates slightly. For a person who sleeps at night and stays awake during the day (no night/shift workers), at what time of day is body temperature the lowest? Be specific for full marks. (2 marks)

7. At what time of day is body temperature highest for a person who sleeps at night and stays awake during the day? Be specific for full marks. (2 marks)

---------- or ----------

Option B: Do You Hear what I Hear?
There are different types of hearing loss. Hearing loss may result from a tumor, from trauma to the ear, or even from noise. Exposure to noise is often measured in decibels. Conduct research (internet or otherwise), regarding noise produced from common activities and then complete the tasks that follow. You may respond on this page.

Sound levels in decibels (dB)



1. Above what decibel level is sound considered dangerous? (2 marks)

   85 dB

2. The thermometer diagram on the left goes from 0 decibels to over 175 decibels. Find three coloured pencils (red, yellow, green) or three highlighters (pink, yellow, green), and show, by colouring the thermometer, the range of decibels that is safe (green), the range of decibels that is borderline (yellow), and the range of decibels that is considered dangerous (red/pink). (3 marks)

   Some internet sources may show "yellow" range up to 110 dB, so accept any reasonable range for the yellow area.

3. On the left side of the thermometer are letters A-G. Correctly match the letter to the sound it represents. (7 marks)

   _C_ Rock concert or ambulance
   _A_ 12-gage shotgun
   _D_ Busy city traffic
   _F_ Whisper
   _G_ Softest sound heard
   _E_ Typical speech
   _B_ Fireworks/firecrackers

4. What is tinnitus? (2 marks)

   Tinnitus is the perception of sound (usually ringing) within the ear when no actual sound is present.

(Response Area for Option A.)

1. Describe, with some detail, the sensations of touching the room-temperature glass after holding the other two glasses for 60 seconds. (4 marks)

   Students should note that the hand originally holding the cold glass feels hot, and the hand holding the hot glass feels cold.

2. What is the commonly used temperature scale in Canada called? (1 mark)

   Celsius

3. What is the commonly used temperature scale in the United States called? (1 mark)

   Fahrenheit

4. What is the average body temperature of an adult in the temperature scale used in Canada? (2 marks)

   The average body temp is 37.0 degrees Celsius.

5. Assessing a patient's temperature is part of a full physical examination. Besides taking a rectal (anal) reading of temperature, list two other sites that may be used for measuring temperature. (2 marks)

   Locations include: mouth, under the arm, ear, vagina, urethra, skin of forehead, over the temporal artery

6. Body temperature fluctuates slightly. For a person who sleeps at night and stays awake during the day (no night/shift workers), at what time of day is body temperature the lowest? Be specific for full marks. (2 marks)

   If students say "at night," give 1 mark. If students say at 4:00 a.m. (give or take), award 2 marks.

7. At what time of day is body temperature highest for a person who sleeps at night and stays awake during the day? Be specific for full marks. (2 marks)

   If students say "during the day," give 1 mark. If students say between 4:00- 6:00 p.m. (give or take), award 2 marks.

End of Assignment Booklet 7
After you have completed all items, submit your work to your marker.

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